

Parsa Poorsistani

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EDUCATION

- **Amirkabir University of Technology** 2020 — 2025
BSc in Computer Science

PUBLICATIONS

- C. Zhao, P. Poorsistani, T. Islam, M. Goudarzi, A. N. Toosi (2025). **GraphFlash: Enabling Fast and Elastic Graph Processing on Serverless Infrastructure**. *46th International Conference on Distributed Computing Systems (ICDCS '26)*
- MA. Mirzaei, P. Poorsistani, SA. Javadi (2024). **LEAD: Latency-Efficient Application Deployment for Microservices Architecture**. *29th International Conference on Automation and Computing (ICAC '24)*

RESEARCH EXPERIENCE

University of Michigan Remote
Research Assistant Under Prof. Ang Chen 8/2024 — 12/2024

- Extended SpotProxy, a censorship circumvention system, to support full Layer 4 protocol coverage through WireGuard integration. Implemented custom packet encapsulation protocol and concurrent architecture in Go to improve system scalability.

University of Melbourne Remote
Research Assistant Under Prof. Adel N. Toosi 4/2024 — 3/2025

- Implemented core optimizations for GraphFlash in Go, a serverless graph processing framework, including selective write-back mechanisms, custom serialization with compression, and message batching. Developed and evaluated a Kubernetes-based infrastructure using Knative for deployment portability.

Amirkabir University of Technology Tehran, Iran
Research Assistant Under Prof. Seyyed Ahmad Javadi 9/2023 — 9/2024

- Developed LEAD, a latency-aware application scheduling system for Kubernetes, resulting in a published paper as co-first author. Designed and implemented a multi-node Kubernetes cluster for experimental validation. Built a custom plugin integrating with Prometheus for real-time rescheduling.

INDUSTRY EXPERIENCE

Cafe Bazaar Tehran, Iran
Software Engineer 10/2024 — 7/2026

- Served as owner and primary maintainer of Bazaar's core Video-on-Demand backend service. Implemented VOD platform features and monetization infrastructure in Go and Python, such as video advertisement delivery and subscription-based services, in collaboration with cross-functional engineering and product teams.
- Contributed to the migration of a Python-based backend into a scalable Go service.
- Built and maintained real-time video analytics pipelines in Go, transforming client playback telemetry into Avro-based Kafka streams for downstream large-scale data processing and business analytics workloads.
- Owned and maintained the team's data-infrastructure, including Spark jobs and Airflow DAGs, ensuring reliability and scalability of the team's YARN cluster for internal analytics and reporting pipelines.
- Contributed to company-wide backend systems beyond the VOD stack, such as search infrastructure and high-traffic client-serving services responsible for content and page delivery across Bazaar applications.

Hamkaran System Tehran, Iran
Software Engineer 3/2023 — 12/2023

- Developed enterprise software features in C# for Iran's leading ERP platform as part of the Manufacturing Team, working closely with product stakeholders to deliver business-critical functionality and operational improvements.

Rahnema College Tehran, Iran
Software Engineer Intern 7/2022 — 10/2022

- Selected among 24 participants from over 3000 applicants through a competitive programming-based admission process for a software engineering internship program sponsored by **Tapsi**.

- Served as back-end developer on a 6-person team building a Fantasy Premier League (FPL) platform, designing backend services and APIs using Node.js, MongoDB, and Redis, along with deployment pipelines.

Seft Educational Solutions

Back-end Developer

Tehran, Iran

7/2021 — 1/2022

- Developed microservice APIs using Django-REST for an educational consulting platform with task management functionality.

TEACHING EXPERIENCE

Amirkabir University of Technology

Teaching Assistant

Tehran, Iran

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| • Data Structures and Algorithms | <i>Fall 2023, Spring 2023</i> |
| • Design of Algorithms | <i>Spring 2023</i> |
| • Advanced Programming | <i>Spring 2023</i> |
| • Compiler Design | <i>Fall 2023</i> |
| • Operating Systems | <i>Spring 2024</i> |
| • Computer Networks | <i>Spring 2024</i> |

TECHNICAL SKILLS

Languages: Go, Python, JavaScript/Node.js, Rust, Bash, Java, C, C#

Tools: Kubernetes, Docker, Django, Kafka, Spark, Airflow, MongoDB, PostgreSQL, Redis, Linux, gRPC, Prometheus, Grafana